

Lecture 39 — Least squares, data fitting, and positive definiteness

Wed Dec 2

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, and Lachniet Ch. 2 (least squares). Software lab is Friday: [Lab 6](#). Related objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A .
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that $A^T A$ is symmetric positive semidefinite, explain $x^T A^T A x = \|Ax\|^2$, and recognize that $A^T A$ is positive definite when A has linearly independent columns.